

# Lab: Differential Expression via RNA-Seq Analysis

Genomic Technologies Workshop 2026 (PLPTH885)

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# Outline

- Differential expression test using DESeq2
- Result visualization

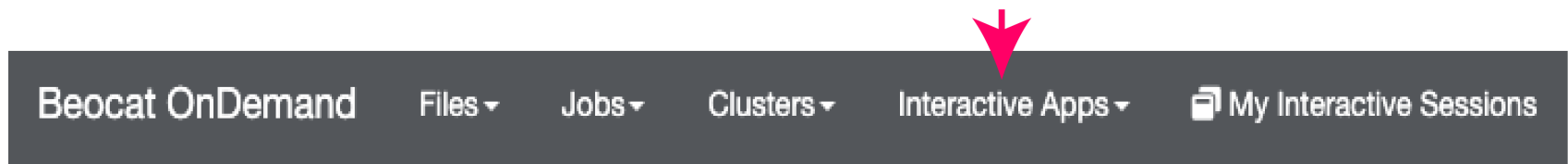
## Course webpage

RNA-seq DE analysis

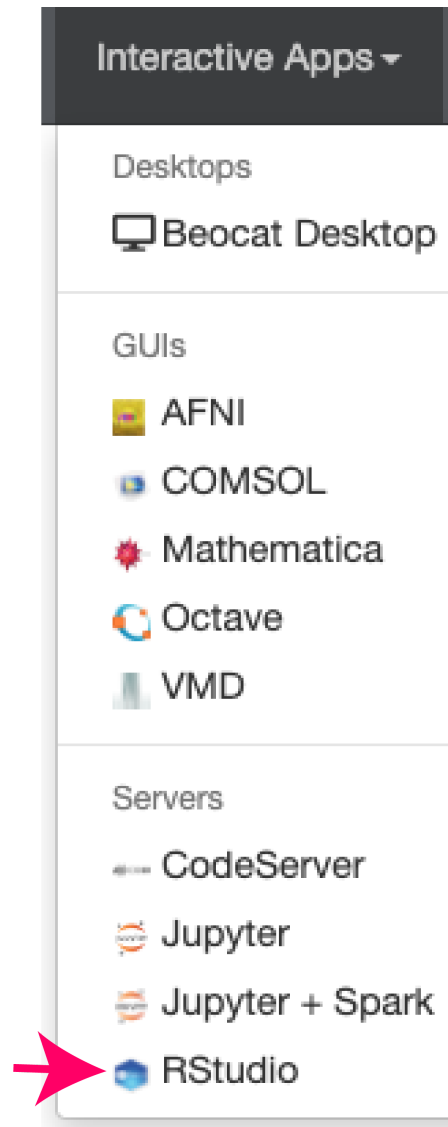
# OnDemand at Beocat

ondemand

login with eID



# Select RStudio



# Request resources

## RStudio

This app will launch [RStudio Server](#) an IDE for [R](#).

### R version

4.3.2 (foss-2023a)



This defines the version of R you want to load.

### Number of hours

24

### Number of cores

1

### Amount of memory

16

The amount of memory (in GB) needed for the whole job

### Job Type

normal



# Connect to RStudio

**RStudio (5655555)**1 node | 1 core | Running

**Host:** >\_wizard24.beocat.ksu.edu

**Created at:** 2022-06-05 13:25:22 CDT

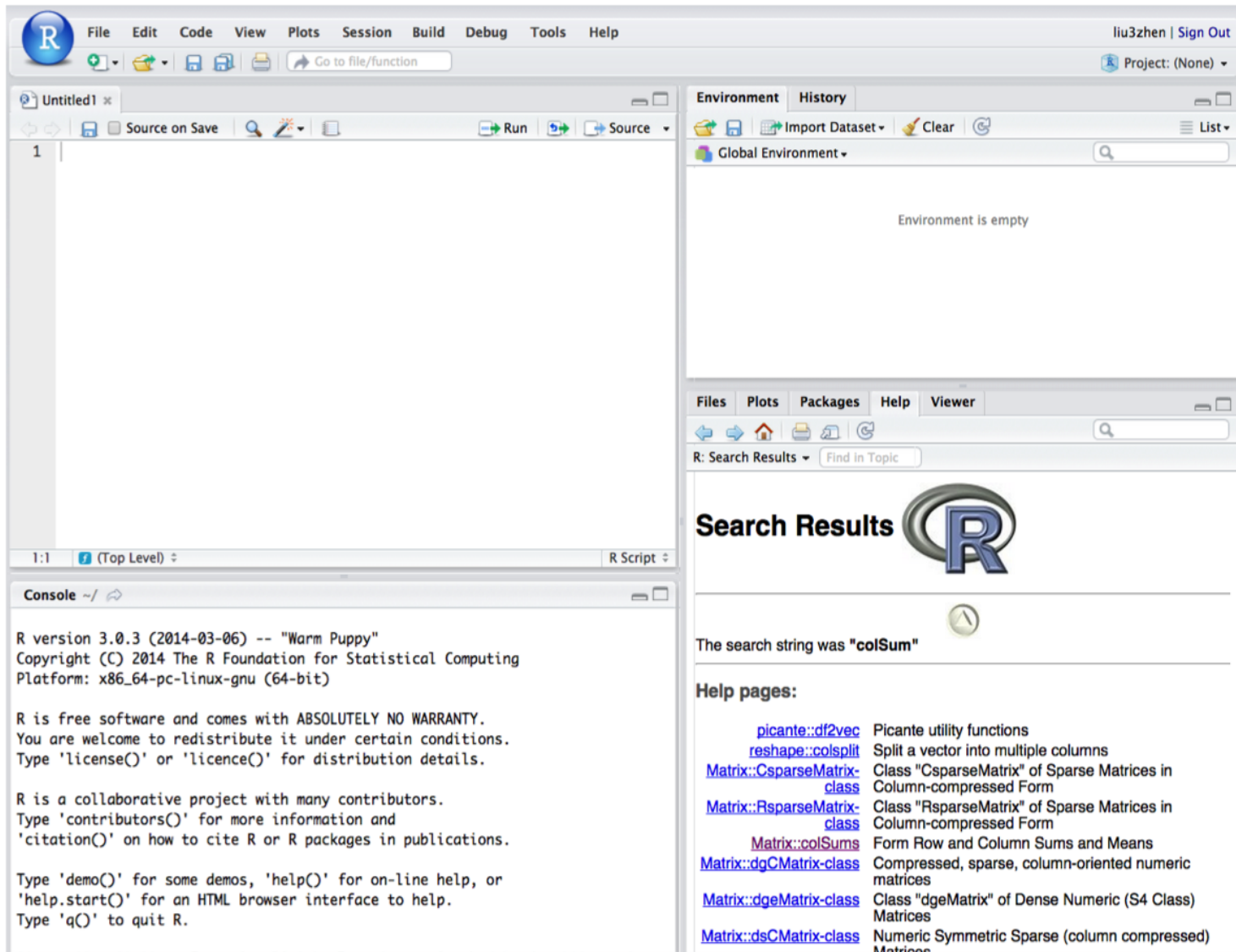
**Time Remaining:** 2 hours and 59 minutes

**Session ID:** 8039a538-a703-42c0-9d6f-47a3a0c0a1c2





# Rstudio interface



# Package installation

## DESeq2 installation

```
if (!require("BiocManager", quietly=T))  
  install.packages("BiocManager")  
  
if (!require("DESeq2", quietly=T))  
  BiocManager::install("DESeq2") # DESeq2
```

## ClusterProfiler

```
if (!require("ggplot2", quietly=T)) {  
  biocmanger_version <- BiocManager::version()  
  if (biocmanger_version == "3.18") {  
    install.packages("https://cran.r-project.org/src/contrib/Archive,  
      repos = NULL, type = "source")  
  } else {  
    install.packages("ggplot2")  
  }  
}  
  
if (!require("clusterProfiler", quietly=T))  
  BiocManager::install("clusterProfiler")
```



# preload modules

```
githubpage <- "https://raw.githubusercontent.com/liu3zhenlab/teaching  
data_url <- paste0(githubpage, "/master/RNA-Seq-Workshop/2026/lab_DE'  
preload_r <- paste0(data_url, "/utils/load.R")  
  
source(preload_r)
```

- panel.cor2
- rnaseq.pca
- normalization

codes

# Read expression data (Read counts per gene)

```
rc_file <- paste0(data_url, "/data/rc.txt")  
grc <- read.delim(rc_file)  
nrow(grc)  # the number of rows/lines
```

[1] 19999

first entry:

| Gene            | trt1 | trt2 | trt3 | ctl1 | ctl2 | ctl3 |
|-----------------|------|------|------|------|------|------|
| Zm00001eb000010 | 283  | 191  | 166  | 308  | 268  | 114  |

last entry:

| Gene            | trt1 | trt2 | trt3 | ctl1 | ctl2 | ctl3 |
|-----------------|------|------|------|------|------|------|
| Zm00001eb222700 | 0    | 0    | 0    | 0    | 0    | 0    |

# RPKM normalization

```
datacols <- c("trt1", "trt2", "trt3",  
             "ctl1", "ctl2", "ctl3")  
lib.sizes <- colSums(grc[, datacols]) # library sizes  
grcn <- normalization(counts = grc,  
                      normcolname = datacols,  
                      libsize = lib.sizes,  
                      methods = "RPM")
```

| Gene            | trt1 | trt2 | trt3 | ctl1 | ctl2 | ctl3 | trt1.RPM | trt2.RPM | trt3.RPM | ctl1.RPM | ctl2.RPM | ctl3.RPM |
|-----------------|------|------|------|------|------|------|----------|----------|----------|----------|----------|----------|
| Zm00001eb000010 | 283  | 191  | 166  | 308  | 268  | 114  | 34.888   | 20.814   | 20.813   | 26.143   | 30.537   | 14.137   |
| Zm00001eb000020 | 264  | 190  | 199  | 302  | 174  | 208  | 32.546   | 20.705   | 24.950   | 25.633   | 19.826   | 25.795   |

# data organization for DESeq2

- count information

```
geneid <- grc$Gene  
in.data <- as.matrix(grc[, 2:7])
```

| trt1 | trt2 | trt3 | ctl1 | ctl2 | ctl3 |
|------|------|------|------|------|------|
| 283  | 191  | 166  | 308  | 268  | 114  |
| 264  | 190  | 199  | 302  | 174  | 208  |
| 0    | 0    | 0    | 0    | 0    | 0    |

## sample names and grouping information (treatment)

```
sample.ids <- colnames(in.data)
treatment <- c("trt", "trt", "trt", "ctl", "ctl", "ctl")
sample.info <- data.frame(row.names=sample.ids, trt=treatment)
```

|      | trt |
|------|-----|
| trt1 | trt |
| trt2 | trt |
| trt3 | trt |
| ctl1 | ctl |
| ctl2 | ctl |
| ctl3 | ctl |

# Differential expression test

## Data organized for DESeq2

```
dds <- DESeqDataSetFromMatrix(countData=in.data,  
                              colData=sample.info,  
                              formula(~trt))
```

```
dds
```

```
class: DESeqDataSet  
dim: 19999 6  
metadata(1): version  
assays(1): counts  
rownames: NULL  
rowData names(0):  
colnames(6): trt1 trt2 ... ctl2 ctl3  
colData names(1): trt
```

## DESeq2 test

```
dds <- DESeq(dds)
```

# DE output

```
res <- results(object = dds)
res <- data.frame(res)
res$Gene <- geneid
res <- res[,c("Gene", "baseMean", "log2FoldChange", "pvalue", "padj")]
nrow(res)
```

```
[1] 19999
```

## DE + normalized data

```
### Merge the normalized result with the DE result  
out <- merge(grcn, res, by = "Gene")  
out <- data.frame(out)
```

| Gene            | trt1 | trt2 | trt3 | ctl1 | ctl2 | ctl3 |
|-----------------|------|------|------|------|------|------|
| Zm00001eb000010 | 283  | 191  | 166  | 308  | 268  | 114  |
| Zm00001eb000020 | 264  | 190  | 199  | 302  | 174  | 208  |

| trt1.RPM | trt2.RPM | trt3.RPM | ctl1.RPM | ctl2.RPM | ctl3.RPM |
|----------|----------|----------|----------|----------|----------|
| 34.888   | 20.814   | 20.813   | 26.143   | 30.537   | 14.137   |
| 32.546   | 20.705   | 24.950   | 25.633   | 19.826   | 25.795   |

| baseMean | log2FoldChange | pvalue    | padj      |
|----------|----------------|-----------|-----------|
| 217.4257 | -0.0277417     | 0.9342927 | 0.9777347 |
| 219.4751 | 0.0048971      | 0.9853634 | 0.9954542 |



## significant gene sets at different FDRs

```
sum(!is.na(out$padj) & out$padj < 0.05)
```

```
[1] 1571
```

### problem

Please revise the code to calculate the number of significant genes with the FDR smaller than 10% and 15%?

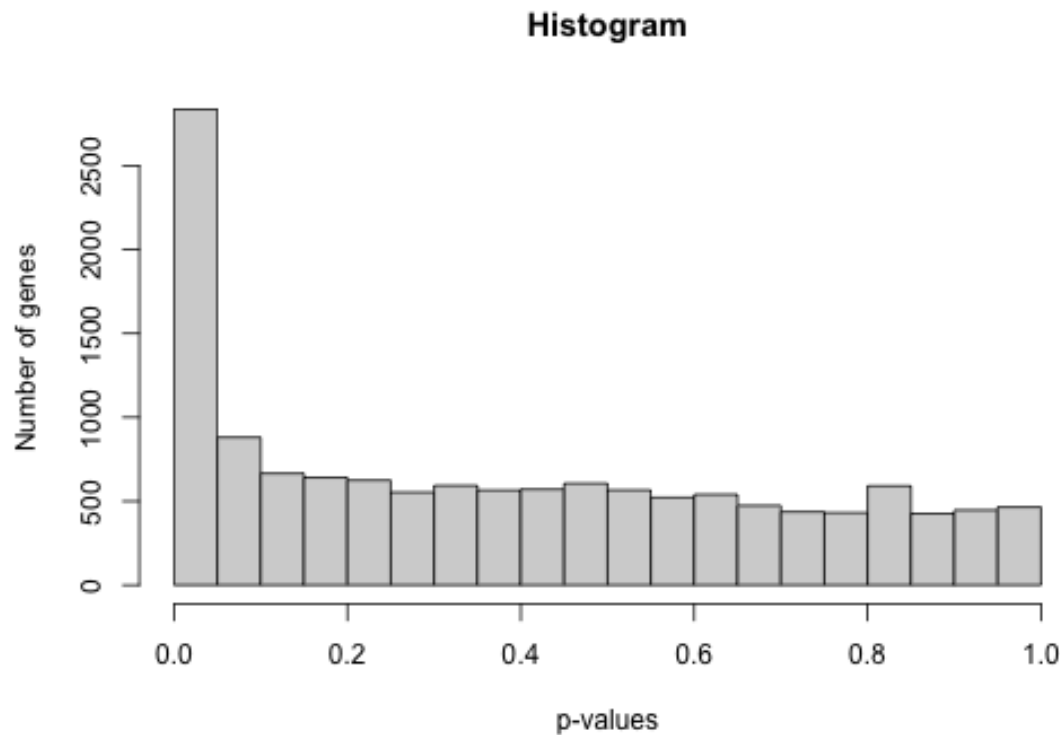
## significantly DEG

```
sig <- out[!is.na(out$padj) & out$padj < 0.05, ]
```

|    | Gene            | trt1.RPM | ctl1.RPM | log2FoldChange | pvalue   | padj      |
|----|-----------------|----------|----------|----------------|----------|-----------|
| 29 | Zm00001eb000370 | 5.917    | 49.399   | -2.7481125     | 0.00e+00 | 0.0000000 |
| 34 | Zm00001eb000420 | 40.313   | 77.409   | -0.8738775     | 1.78e-05 | 0.0003425 |

# p-value histogram

```
pvals <- out$pvalue  
hist(pvals, main="Histogram",  
      xlab="p-values",ylab="Number of genes")
```

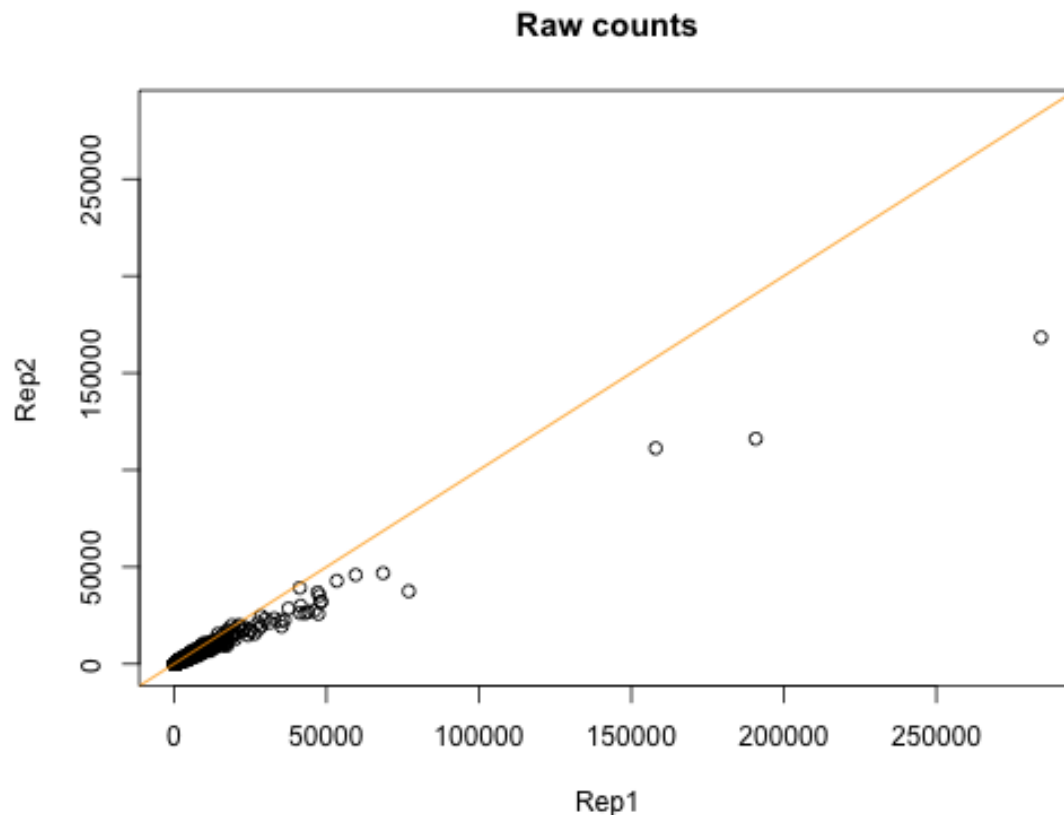


## problem

Please modify the plot code to change the figure title to "DE"

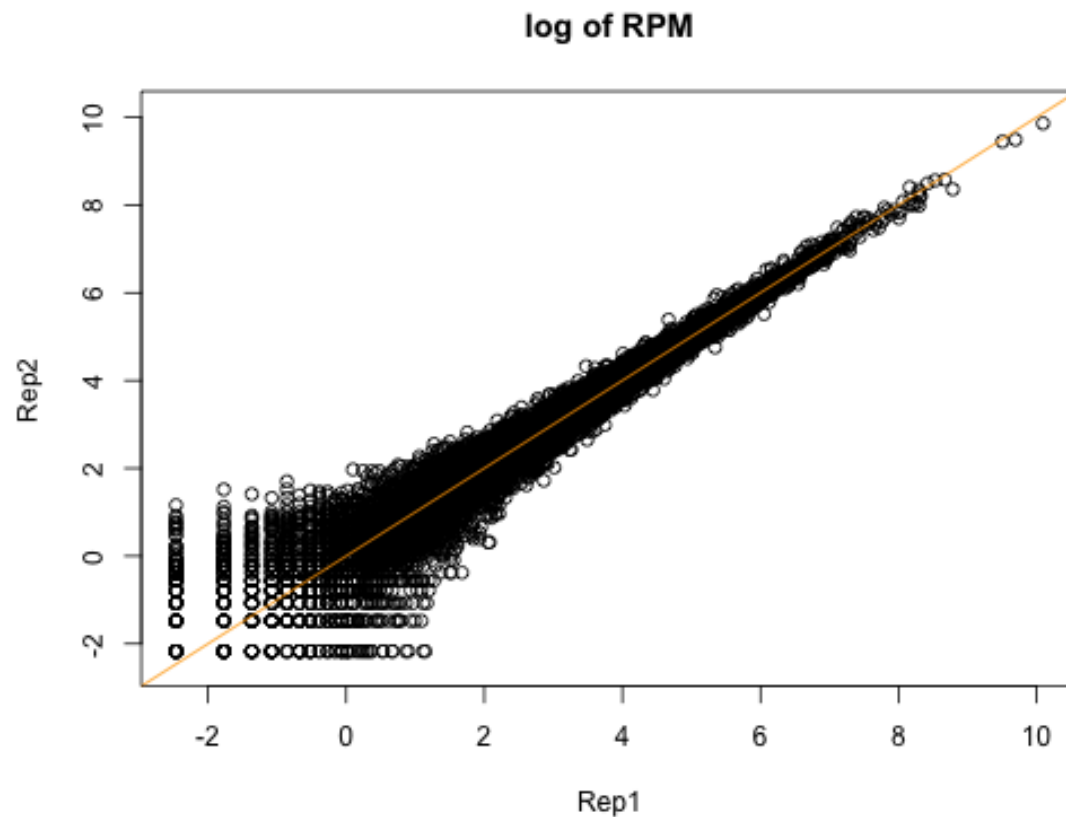
## scatter plot - raw counts

```
x.raw <- out$ctl1; y.raw <- out$ctl2  
drange <- range(x.raw, y.raw, finite=T)  
# plot a scatter plot between two samples  
plot(x.raw, y.raw, main="Raw counts", xlab="Rep1",  
      ylab="Rep2", xlim=drange, ylim=drange)  
abline(a = 0, b = 1, col = "orange")
```



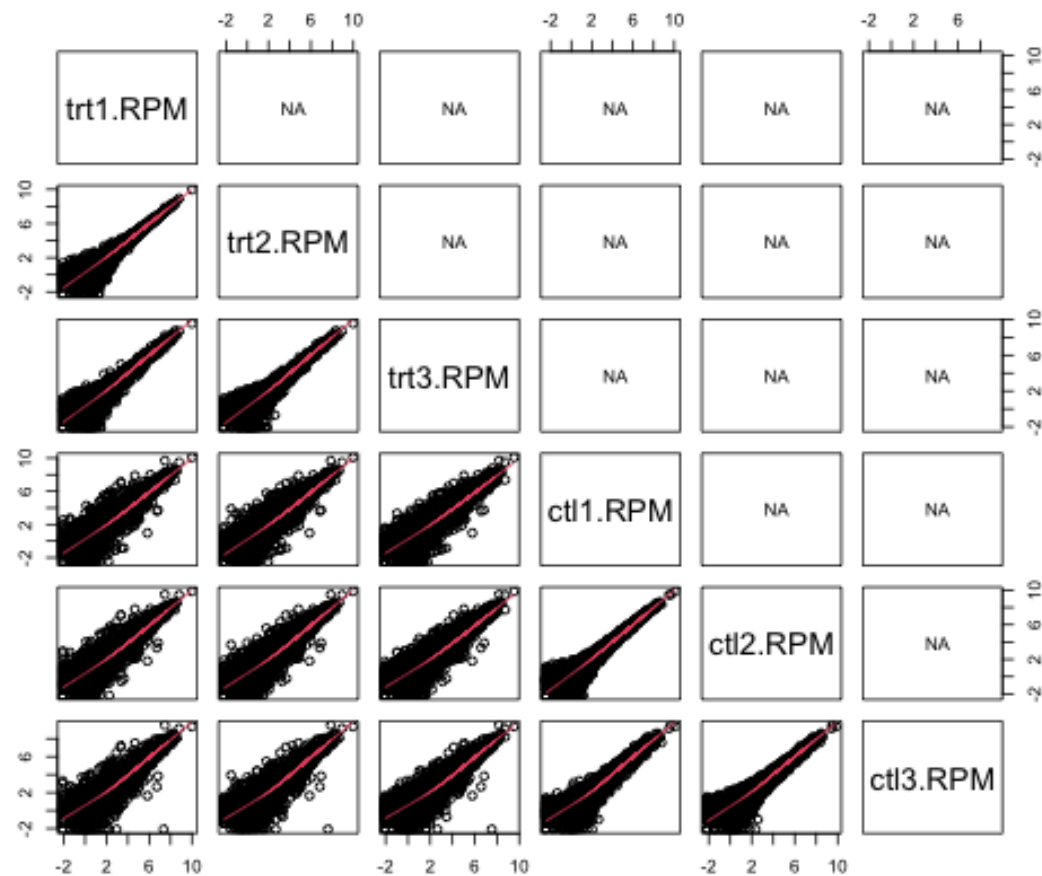
## scatter plot - RPM

```
x <- log(out$ctl1.RPM); y <- log(out$ctl2.RPM)
drange <- range(x, y, finite=T)
plot(x, y, main="log of RPM", xlab="Rep1",
      ylab="Rep2", xlim=drange, ylim=drange)
abline(a = 0, b = 1, col="orange")
```



# pair-wise scatter plots

```
logrpm <- log(out[, 8:13])  
pairs(logrpm, lower.panel=panel.smooth, upper.panel=panel.cor2)
```

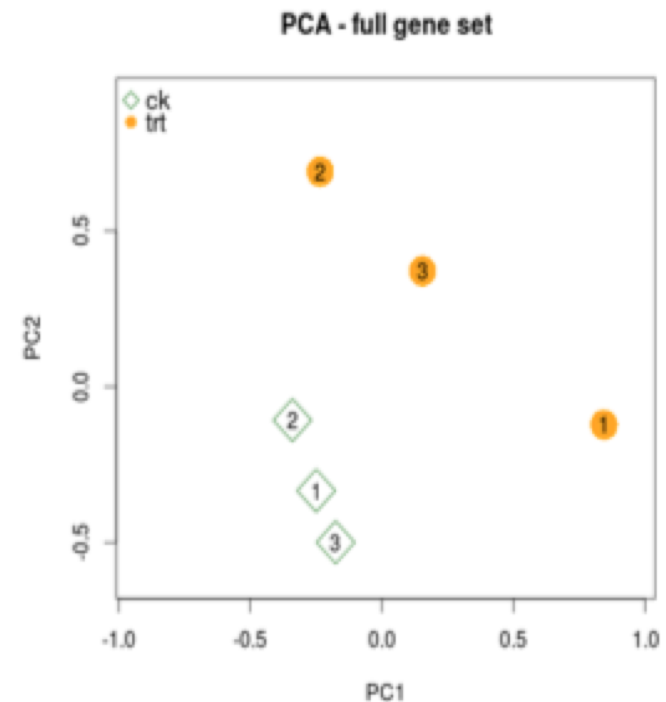


# Principal Component Analysis (PCA)

PCA is a mathematical algorithm that reduces the dimensionality of the data while retaining most of the variation in the data set.

| Control |      |      |      | Treatment |      |      |
|---------|------|------|------|-----------|------|------|
| Gene    | Rep1 | Rep2 | Rep3 | Rep1      | Rep2 | Rep3 |
| 1       | 2679 | 2360 | 2573 | 2563      | 3398 | 3012 |
| 2       | 177  | 161  | 171  | 154       | 137  | 152  |
| 3       | 381  | 371  | 397  | 541       | 723  | 635  |
| ...     |      |      |      |           |      |      |
| 20000   | 990  | 1073 | 1236 | 850       | 672  | 859  |

Normalized and standardized data



# function / module

You can write your own function:

```
fun_name <- function (...) { ... }
```

```
gpa_improve <- function(gpa, rate) {  
  ### gpa: a numeric vector for GPAs  
  ### rate: percentage for the improvement  
  new.gpa <- gpa * (1 + rate)  
  new.gpa[new.gpa > 4] <- 4  
  return(new.gpa)  
}
```

```
### running the function  
our.gpa <- c(3.8, 3.3, 2.8, 3.1)  
gpa_improve(our.gpa, 0.1)
```

```
[1] 4.00 3.63 3.08 3.41
```

```
gpa_improve(our.gpa, 0.2)
```

```
[1] 4.00 3.96 3.36 3.72
```



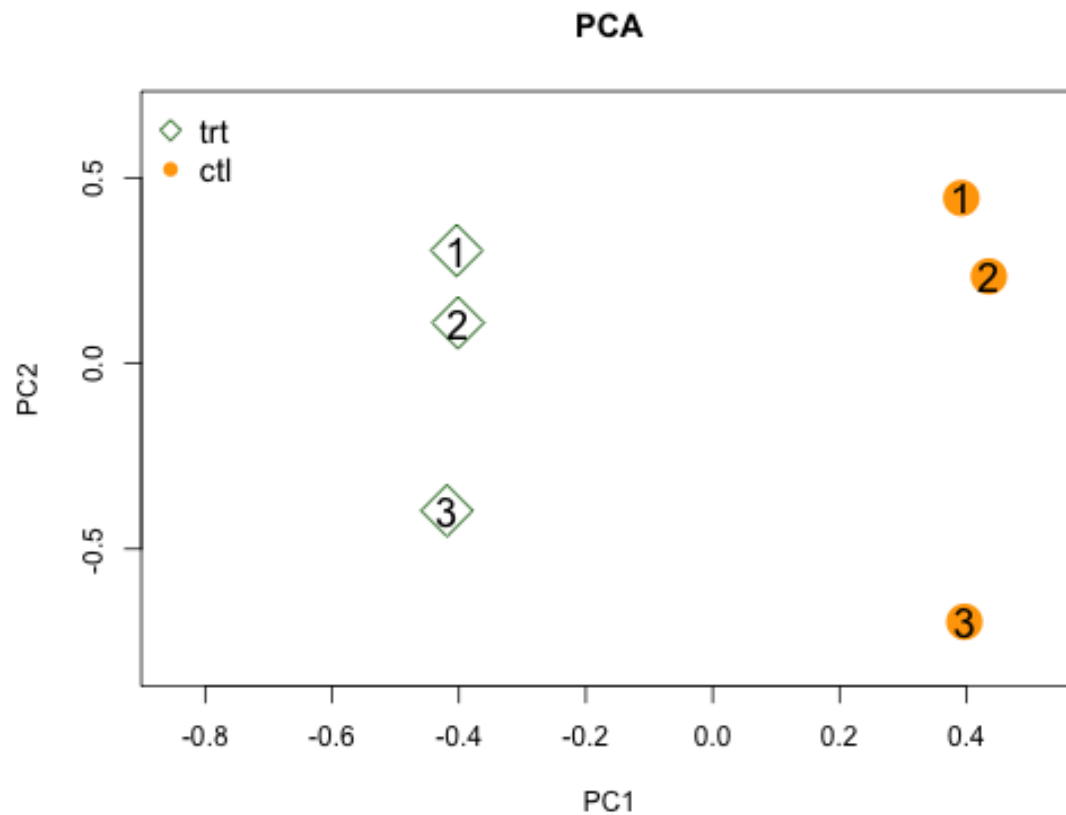
# PCA function

## principal component analysis and plotting

```
rnaseq.pca <- function(norm.data,  
  norm.feature="RPM",  
  group.feature,  
  title="",  
  shape.code=NULL,  
  mean.cutoff=0.1,  
  colors=NULL,  
  scaling=T, ...) {  
  ...  
}
```

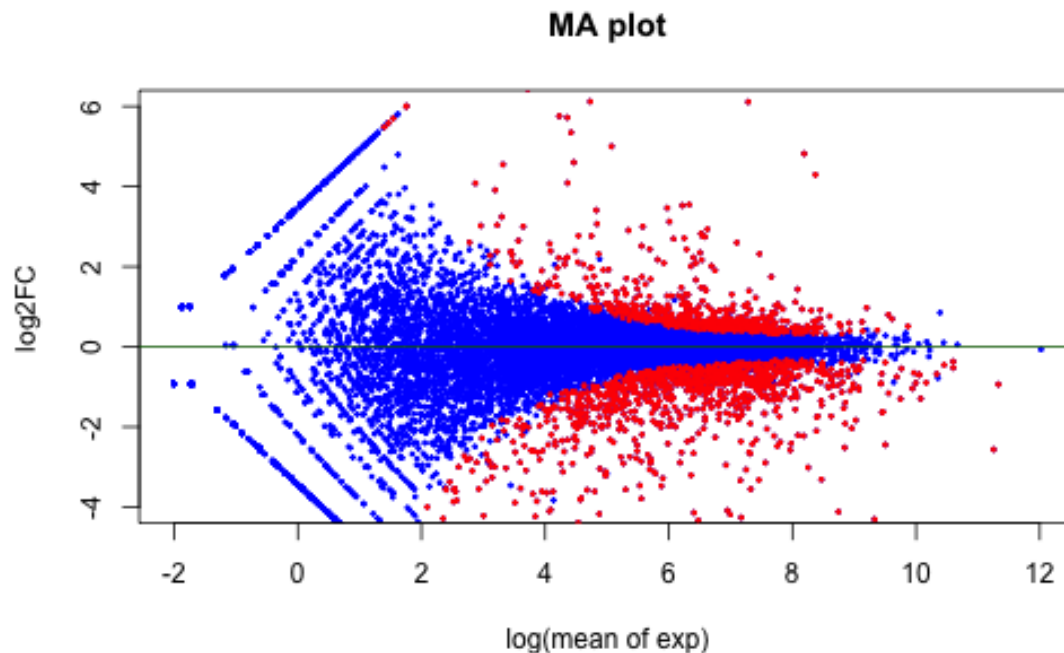
# PCA plotting

```
rnaseq.pca(out, norm.feature="RPM",  
  group.feature=c("trt", "ctl"), shape.code=c(5, 16),  
  mean.cutoff=0.1, cex=3, scaling = T,  
  title="PCA", colors=c("dark green","orange"))
```



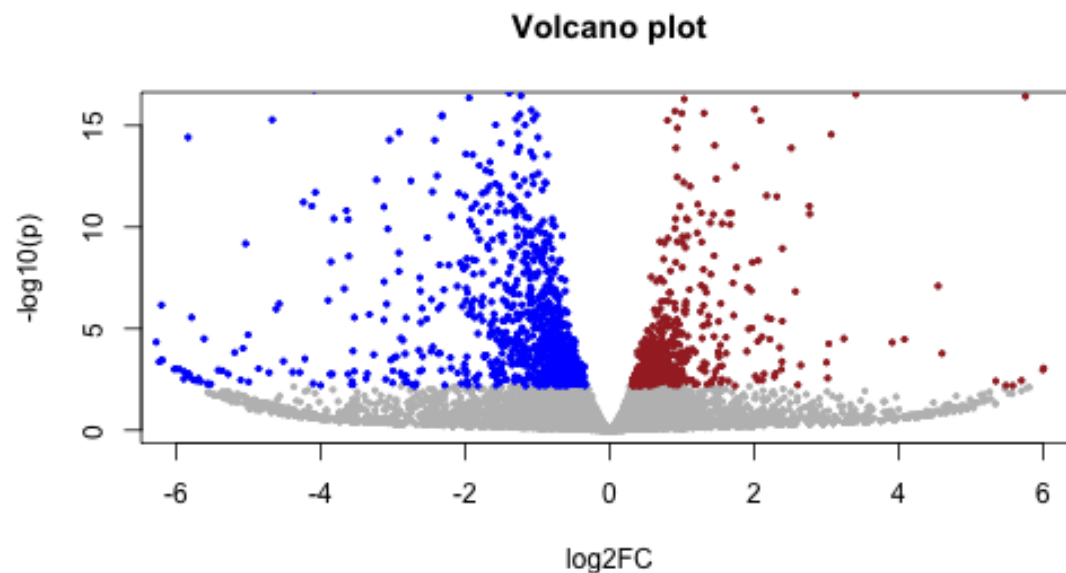
# MA plot

```
aval <- log(out$baseMean)
mval <- out$log2FoldChange
sig.set <- which(out$padj < 0.05)
plot(aval, mval, main="MA plot", xlab="log(mean of exp)",
     ylab="log2FC", ylim=c(-4,6),pch=19,cex=0.3,col="blue")
# add points and a horizontal line
points(aval[sig.set], mval[sig.set], pch=19,cex=0.3,col="red")
abline(h = 0, cex = 1, col = "dark green")
```



# Volcano plot

```
log2val <- out$log2FoldChange # log2 fold change
mlogP <- -log10(out$pvalue) # transformed p-values
### plot
plot(log2val, mlogP, main="Volcano plot", xlab="log2FC",
     ylab="-log10(p)", pch=19, cex=0.4, col="grey", xlim=c(-6,6), ylim=c(0,16))
### highlight
up <- which(out$padj < 0.05 & log2val > 0) # up
dn <- which(out$padj < 0.05 & log2val < 0) # down
points(log2val[up], mlogP[up], pch=19, cex=0.4, col="brown")
points(log2val[dn], mlogP[dn], pch=19, cex=0.4, col="blue")
```



# Gene Ontology enrichment - package installation

clusterProfiler

```
library(GO.db)  
library(ggplot2)  
library(clusterProfiler)
```

# GO database

```
# GO DB (2 columns)
go <- read.delim("https://raw.githubusercontent.com/liu3zhenlab/teach
colnames(go) <- c("Gene", "GO")
go <- go[, c("GO", "Gene")] # switch gene and GO
is_go_valid <- go$GO %in% keys(GOTERM)
go <- go[is_go_valid, ]
```

| GO         | Gene            |
|------------|-----------------|
| GO:0003690 | Zm00001eb000010 |
| GO:0006353 | Zm00001eb000010 |
| GO:0006355 | Zm00001eb000010 |
| GO:0009507 | Zm00001eb000010 |
| GO:0032502 | Zm00001eb000010 |
| GO:0000712 | Zm00001eb000020 |
| GO:0005634 | Zm00001eb000020 |
| GO:0005737 | Zm00001eb000020 |

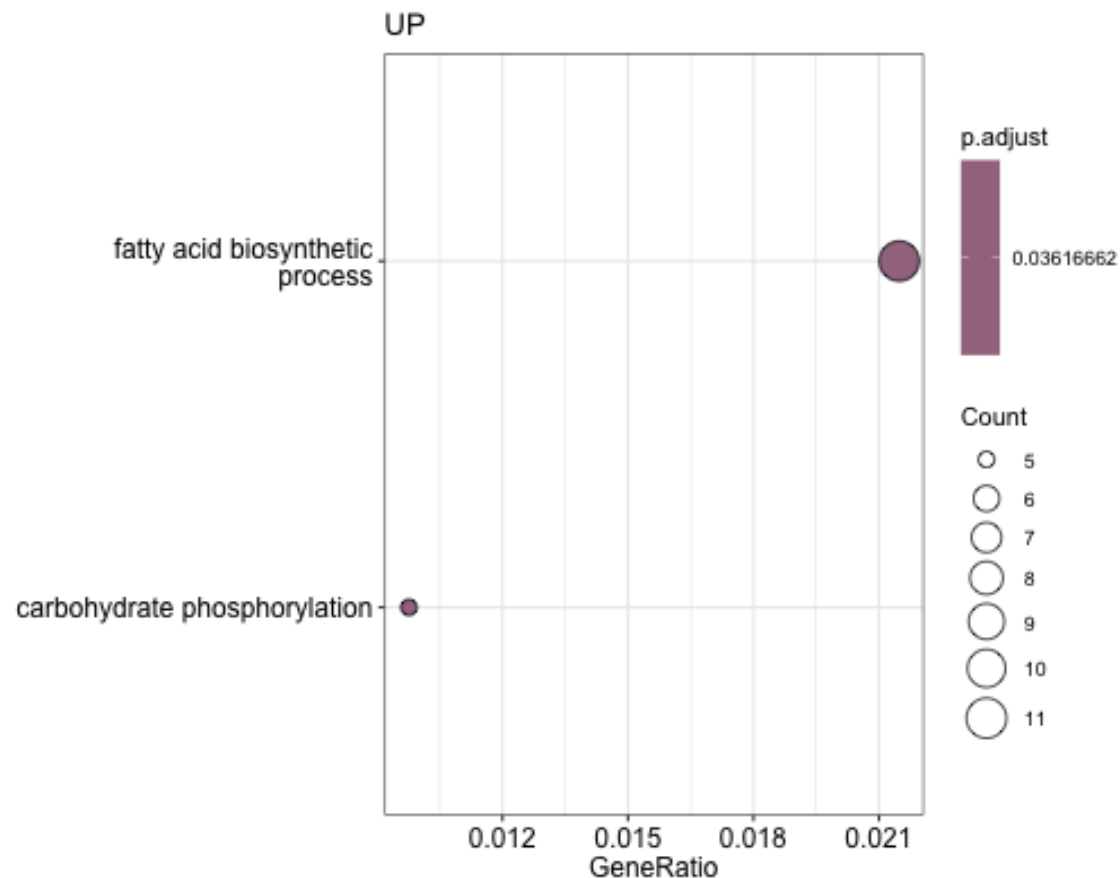
# DEG (up- and down-regulated)

```
up_gene <- out$Gene[up]  
dn_gene <- out$Gene[dn]
```

In total, 572 and 999 up- and down-regulated genes were identified, respectively.

# GO enrichment test

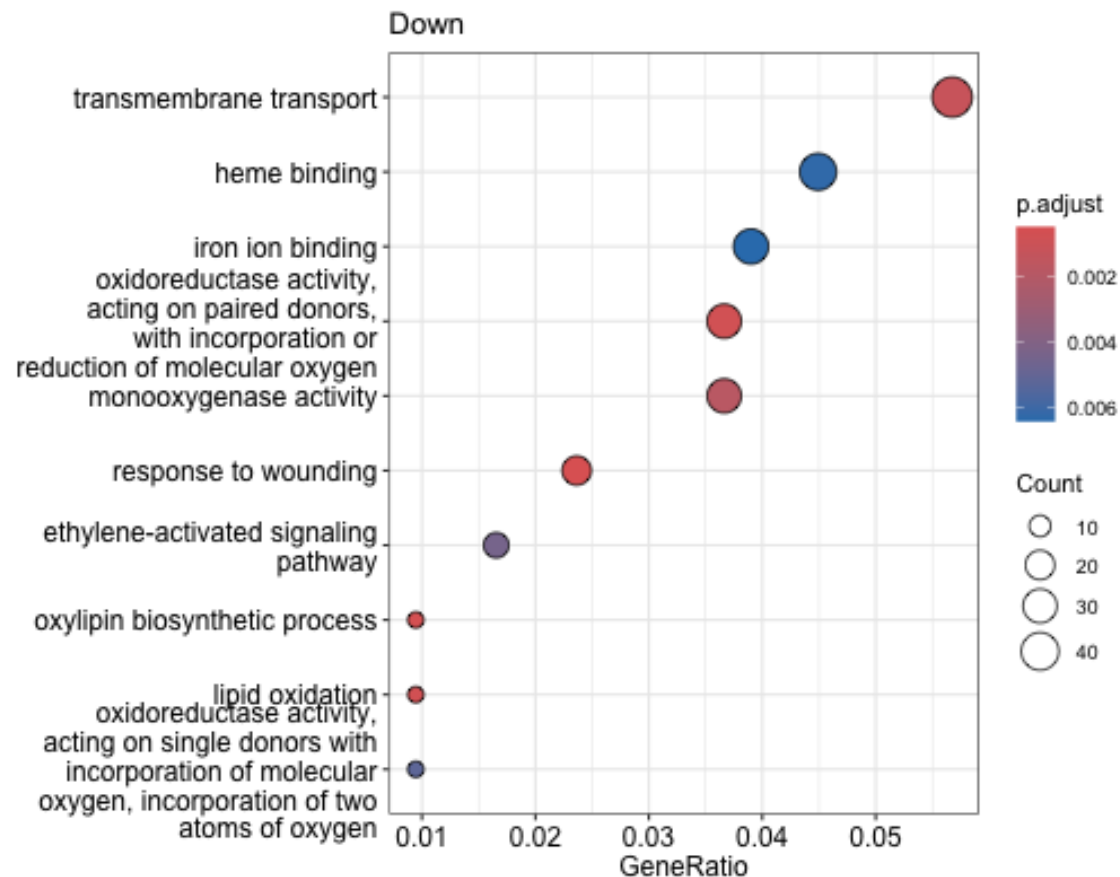
```
up_enrich <- enricher(up_gene, universe=out$Gene, TERM2GENE=go)
up_enrich@result$Description <- Term(GOTERM[up_enrich@result$ID])
up_dp <- dotplot(up_enrich, showCategory=10, title="UP", x="GeneRatio")
up_dp + aes(y = Description)
```





# Significant down-regulated

```
dn_enrich <- enricher(gene=dn_gene, universe=out$Gene, TERM2GENE=go)
dn_enrich@result$Description <- Term(GOTERM[dn_enrich@result$ID])
dn_dp<-dotplot(dn_enrich,showCategory=10,title="Down",x="GeneRatio")
dn_dp + aes(y = Description)
```



# Summary of the analyzing procedure

1. Read counts per gene
2. DE analysis based on the experimental design
3. Examine results (p-value distribution, number of significant genes)
4. Gene Ontology enrichment test

DESeq2 tutorial

# Contact information

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# Bioinformatics Applications

PLPTH813, Spring 2027